GEN211 Probability and Statistics

Lecture 3: Probability Rules

Lecture contents

- Basic concepts
- Introduction to probability theory
- Mutually exclusive events and addition/multiplication rules

ELSEWEDY UNIVERSITY OF TECHNOLOGY
- POLYTECHNIC OF EGYPT -

Basic Concepts

Basic concepts

A **probability experiment** is a chance process that leads to well-defined results called outcomes.

An **outcome** is the result of a single trial of a probability experiment.

A **sample space** is the set of all possible outcomes of a probability experiment.

Some sample spaces for various probability experiments are shown here.

Experiment	Sample space
Toss one coin	Head, tail
Roll a die	1, 2, 3, 4, 5, 6
Answer a true/false question	True, false
Toss two coins	Head-head, tail-tail, head-tail, tail-head

Basic concepts

An **event** consists of a set of outcomes of a probability experiment.

An event can be one outcome or more than one outcome. For example, if a die is rolled and a 6 shows, this result is called an *outcome,* since it is a result of a single trial. An event with one outcome is called a **simple event.** The event of getting an odd number when a die is rolled is called a **compound event,** since it consists of three outcomes or three simple events. In general, a compound event consists of two or more outcomes or simple events.

Equally likely events are events that have the same probability of occurring.

Example 1: Sample space

Find the sample space for rolling two dice.

SOLUTION

Since each die can land in six different ways, and two dice are rolled, the sample space can be presented by a rectangular array, as shown. The sample space is the list of pairs of numbers in the chart.

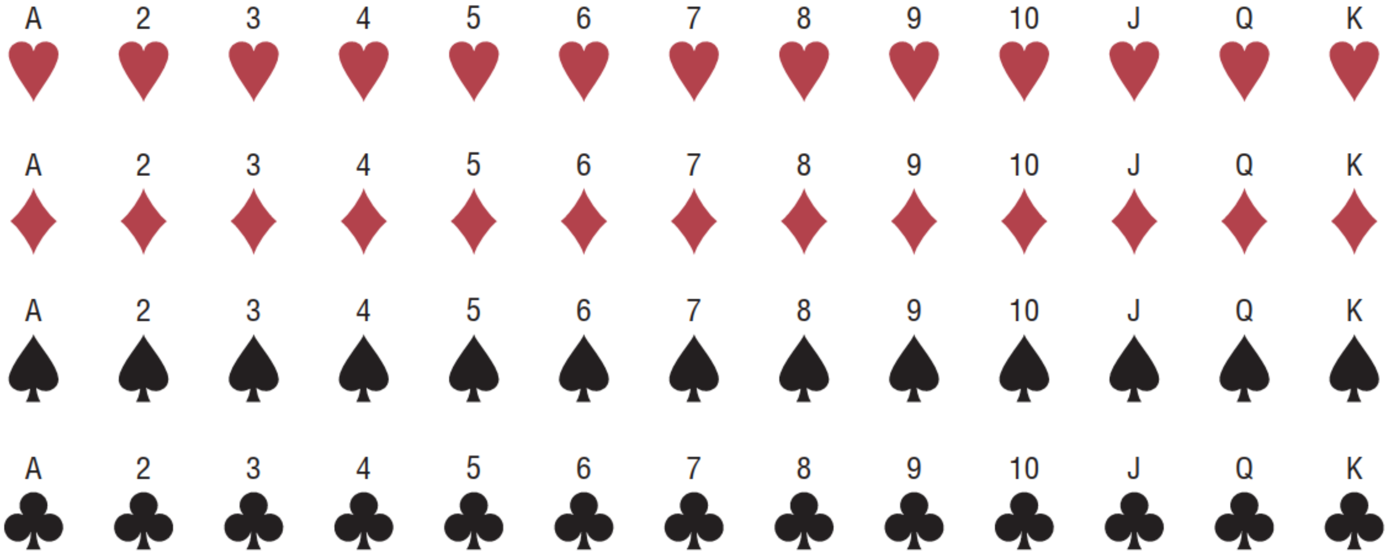
	Die 2					
Die 1	1	2	3	4	5	6
1	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
2	(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
3	(3, 1)	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
4	(4, 1)	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
5	(5, 1)	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
6	(6, 1)	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

Example 2: Sample space

Find the sample space for drawing one card from an ordinary deck of cards.

SOLUTION

Since there are 4 suits (hearts, clubs, diamonds, and spades) and 13 cards for each suit (ace through king), there are 52 outcomes in the sample space.



Example 3: Tree diagram

A **tree diagram** is a device consisting of line segments emanating from a starting point and also from the outcome point. It is used to determine all possible outcomes of a probability experiment.

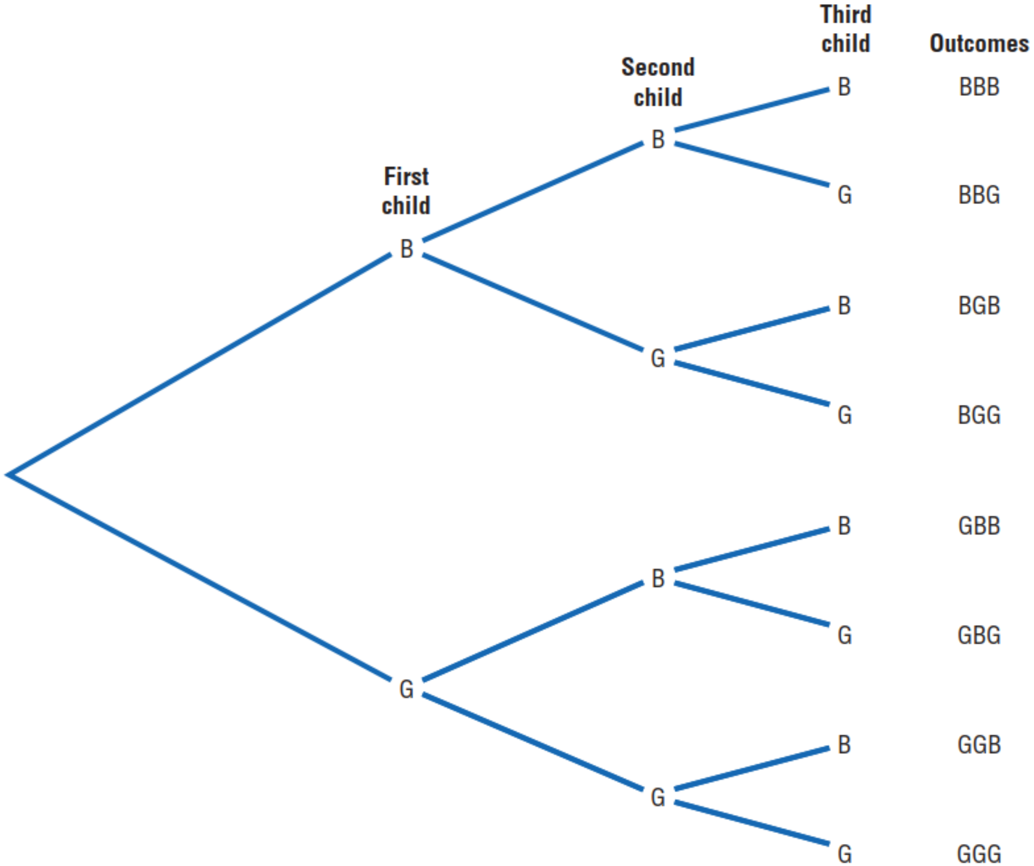
Use a tree diagram to find the sample space for the gender of three children in a family.

SOLUTION

Since there are two possibilities (boy or girl) for the first child, draw two branches from a starting point and label one B and the other G. Then if the first child is a boy, there are two possibilities for the second child (boy or girl), so draw two branches from B and label one B and the other G. Do the same if the first child is a girl. Follow the same procedure for the third child.

Example 3: Tree diagram

SOLUTION



ELSEWEDY UNIVERSITY OF TECHNOLOGY
- POLYTECHNIC OF EGYPT -

Introduction to Probability Theory

Classical probability

Classical probability assumes that all outcomes in the sample space are equally likely to occur.

Formula for Classical Probability

The probability of any event E is

$$\frac{\text{Number of outcomes in } E}{\text{Total number of outcomes in the sample space}}$$

This probability is denoted by

$$P(E) = \frac{n(E)}{n(S)}$$

where $n(E)$ is the number of outcomes in E and $n(S)$ is the number of outcomes in the sample space S.

Examples 4, 5: Classical probability, revisit slides 7-9

Find the probability of getting a red face card when randomly drawing a card from an ordinary deck.

SOLUTION

There are 52 cards in an ordinary deck of cards, and there are 6 red face cards (jack, queen, and king of hearts and jack, queen, and king of diamonds). Hence, the probability of getting a red face card is

$$\frac{6}{52} = \frac{3}{26} \approx 0.115$$

If a family has three children, find the probability that exactly two of the three children are girls.

SOLUTION

The sample space for the gender of the children for a family with three children has eight outcomes, that is, BBB, BBG, BGB, GBB, GGG, GGB, GBG, and BGG. Since there are three ways to have two girls, namely, GGB, GBG, and BGG, $P(\text{two girls}) = \frac{3}{8}$.

Example 6: Classical probability, revisit slide 7

A card is drawn from an ordinary deck. Find the probability of getting

a. A king

b. The 4 of spades

c. A face card (jack, queen, or king)

d. A red card

e. A club

Example 6: Classical probability

SOLUTION

a. There are 4 kings in the event E and a total of 52 cards; hence,

$$P(\text{king}) = \frac{4}{52} = \frac{1}{13} \approx 0.077$$

b. Since there is only one 4 of spades, the probability is

$$P(4\spadesuit) = \frac{1}{52} \approx 0.019$$

c. There are 3 face cards for each suit, and there are 4 suits (hearts, clubs, diamonds, and spades). So there are 12 face cards; hence,

$$P(\text{face card}) = \frac{12}{52} = \frac{3}{13} \approx 0.231$$

d. There are 26 red cards: 13 diamonds and 13 hearts. Hence,

$$P(\text{red card}) = \frac{26}{52} = \frac{1}{2} = 0.5$$

e. There are 13 clubs, so the probability of selecting a club is

$$P(\clubsuit) = \frac{13}{52} = \frac{1}{4} = 0.25$$

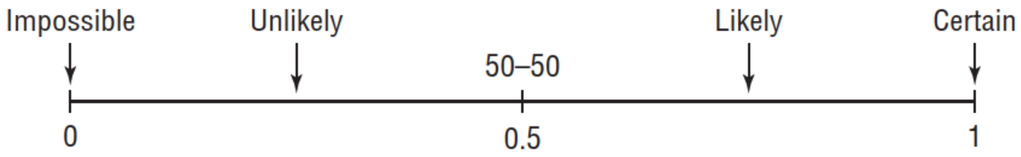
Probability rules

Probability Rules

1. The probability of any event E is a number (either a fraction or decimal) between and including 0 and 1. This is denoted by $0 \leq P(E) \leq 1$.
2. The sum of the probabilities of all the outcomes in a sample space is 1.
3. If an event E cannot occur (i.e., the event contains no members in the sample space), its probability is 0.
4. If an event E is certain, then the probability of E is 1.

Probability rules

Rule 1 states that probability values range from 0 to 1. When the probability of an event is close to 0, its occurrence is highly unlikely. When the probability of an event is near 0.5, there is about a 50-50 chance that the event will occur; and when the probability of an event is close to 1, the event is highly likely to occur.



Rule 2 can be illustrated by the example of rolling a single die. Each outcome in the sample space has a probability of $\frac{1}{6}$, and the sum of the probabilities of all the outcomes is 1, as shown.

Outcome	1		2		3		4		5		6
Probability	$\frac{1}{6}$		$\frac{1}{6}$		$\frac{1}{6}$		$\frac{1}{6}$		$\frac{1}{6}$		$\frac{1}{6}$
Sum	$\frac{1}{6}$	$+$	$\frac{1}{6}$	$+$	$\frac{1}{6}$	$+$	$\frac{1}{6}$	$+$	$\frac{1}{6}$	$+$	$\frac{1}{6} = \frac{6}{6} = 1$

Examples 7, 8: Illustration of rules 3 and 4 of the probability

When a single die is rolled, find the probability of getting a 9.

SOLUTION

Since the sample space is 1, 2, 3, 4, 5, and 6, it is impossible to get a 9. Hence, the probability is $P(9) = \frac{0}{6} = 0$.

When a single die is rolled, what is the probability of getting a number less than 7?

SOLUTION

Since all outcomes—1, 2, 3, 4, 5, and 6—are less than 7, the probability is

$$P(\text{number less than } 7) = \frac{6}{6} = 1$$

The event of getting a number less than 7 is certain.

Example 9: Complementary events

The **complement of an event** *E* is the set of outcomes in the sample space that are not included in the outcomes of event *E*. The complement of *E* is denoted by $\bar{E}$ (read "*E* bar").

Find the complement of each event:

a. Selecting a month that has 31 days

b. Selecting a day of the week that begins with the letter T

c. Rolling two dice and getting a sum that is an odd number

d. Selecting a letter of the alphabet that is used as a vowel or consonant

Example 9: Complementary events

SOLUTION

a. Select a month that has fewer than 31 days, i.e., February, April, June, September, and November.

b. Select a day of the week that does not begin with a T, i.e., Sunday, Monday, Wednesday, Friday, and Saturday.

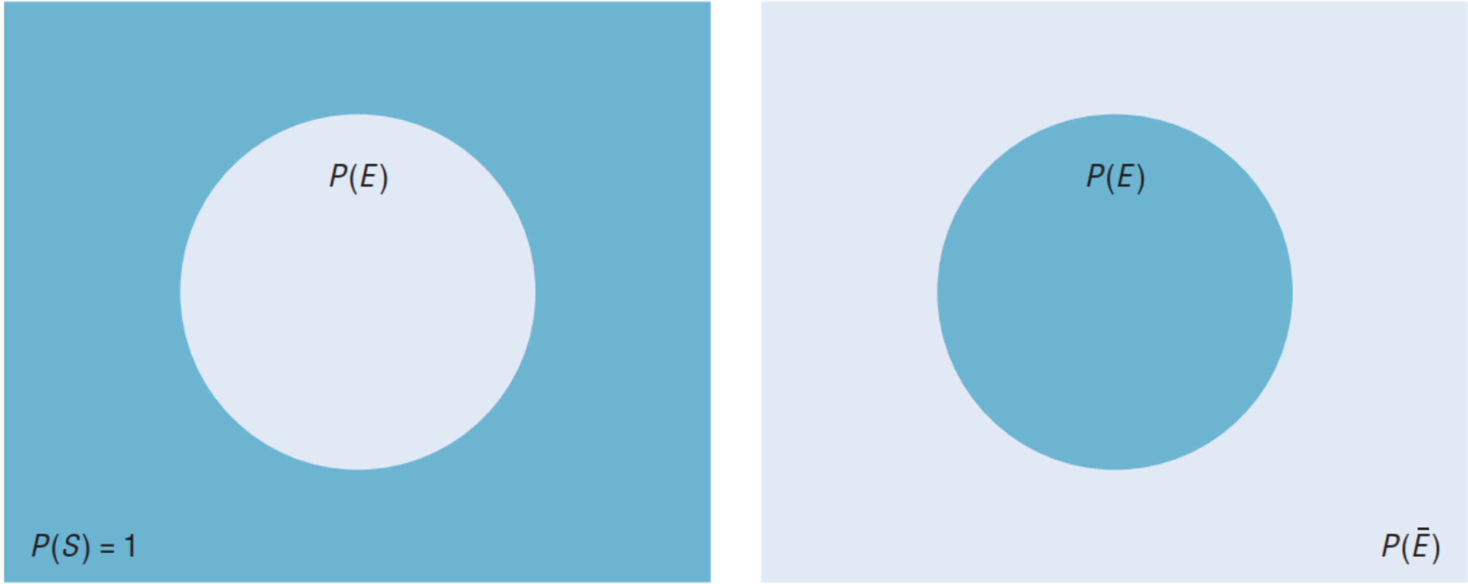
c. Rolling two dice and getting a sum that is an even number, i.e., a sum of 2, 4, 6, 8, 10, or 12.

d. Since y is the only letter that is used as a vowel or a consonant, the complement is selecting a letter other than y.

Probability of complementary events and Venn diagrams

Rule for Complementary Events

$$P(\overline{E}) = 1 - P(E) \qquad \text{or} \qquad P(E) = 1 - P(\overline{E}) \qquad \text{or} \qquad P(E) + P(\overline{E}) = 1$$



(a) Simple probability **(b)** $P(\bar{E}) = 1 - P(E)$

Example 10: Complementary probability

In a study, it was found that 23% of the people surveyed said that vanilla was their favorite flavor of ice cream. If a person is selected at random, find the probability that the person's favorite flavor of ice cream is not vanilla.

SOLUTION

$$\begin{aligned} P(\text{not vanilla}) &= 1 - P(\text{vanilla}) \\ &= 1 - 0.23 = 0.77 = 77\% \end{aligned}$$

Empirical (Relative frequency) probability

The difference between classical and **empirical probability** is that classical probability assumes that certain outcomes are equally likely (such as the outcomes when a die is rolled), while empirical probability relies on actual experience to determine the likelihood of outcomes. In empirical probability, one might actually roll a given die 6000 times, observe the various frequencies, and use these frequencies to determine the probability of an outcome.

Formula for Empirical Probability

Given a frequency distribution, the probability of an event being in a given class is

$$P(E) = \frac{\text{frequency for the class}}{\text{total frequencies in the distribution}} = \frac{f}{n}$$

This probability is called *empirical probability* and is based on observation.

Example 11: Empirical (Relative frequency) probability

In a sample of 50 people, 21 had type O blood, 22 had type A blood, 5 had type B blood, and 2 had type AB blood. Set up a frequency distribution and find the following probabilities.

a. A person has type O blood.

b. A person has type A or type B blood.

c. A person has neither type A nor type O blood.

d. A person does not have type AB blood.

Example 11: Empirical (Relative frequency) probability

SOLUTION

Type	Frequency
A	22
B	5
AB	2
O	21
	Total 50

a. $P(\mathrm{O}) = \dfrac{f}{n} = \dfrac{21}{50}$

b. $P(\mathrm{A \text{ or } B}) = \dfrac{22}{50} + \dfrac{5}{50} = \dfrac{27}{50}$

(Add the frequencies of the two classes.)

c. $P(\text{neither A nor O}) = \dfrac{5}{50} + \dfrac{2}{50} = \dfrac{7}{50}$

(Neither A nor O means that a person has either type B or type AB blood.)

d. $P(\text{not AB}) = 1 - P(\mathrm{AB}) = 1 - \dfrac{2}{50} = \dfrac{48}{50} = \dfrac{24}{25}$

(Find the probability of not AB by subtracting the probability of type AB from 1.)

Example 12: Empirical (Relative frequency) probability

Hospital records indicated that knee replacement patients stayed in the hospital for the number of days shown in the distribution.

Number of days stayed	Frequency
3	15
4	32
5	56
6	19
7	5
	127

Find these probabilities.

a. A patient stayed exactly 5 days.

b. A patient stayed fewer than 6 days.

c. A patient stayed at most 4 days.

d. A patient stayed at least 5 days.

Example 12: Empirical (Relative frequency) probability

SOLUTION

$$a.\ P(5) = \frac{56}{127}$$

$$b.\ P(\text{fewer than 6 days}) = \frac{15}{127} + \frac{32}{127} + \frac{56}{127} = \frac{103}{127}$$

(Fewer than 6 days means 3, 4, or 5 days.)

$$c.\ P(\text{at most 4 days}) = \frac{15}{127} + \frac{32}{127} = \frac{47}{127}$$

(At most 4 days means 3 or 4 days.)

$$d.\ P(\text{at least 5 days}) = \frac{56}{127} + \frac{19}{127} + \frac{5}{127} = \frac{80}{127}$$

(At least 5 days means 5, 6, or 7 days.)

Subjective probability

The third type of probability is called *subjective probability*. **Subjective probability** uses a probability value based on an educated guess or estimate, employing opinions and inexact information.

In subjective probability, a person or group makes an educated guess at the chance that an event will occur. This guess is based on the person's experience and evaluation of a solution. For example, a sportswriter may say that there is a 70% probability that the Pirates will win the pennant next year. A physician might say that, on the basis of her diagnosis, there is a 30% chance the patient will need an operation. A seismologist might say there is an 80% probability that an earthquake will occur in a certain area. These are only a few examples of how subjective probability is used in everyday life.

All three types of probability (classical, empirical, and subjective) are used to solve a variety of problems in business, engineering, and other fields.

Mutually Exclusive Events and Addition/Multiplication Rules

Example 13: Mutually exclusive events

Two events are **mutually exclusive events** or **disjoint events** if they cannot occur at the same time (i.e., they have no outcomes in common).

Determine whether the two events are mutually exclusive. Explain your answer.

a. Randomly selecting a female student
Randomly selecting a student who is a junior

b. Randomly selecting a person with type A blood
Randomly selecting a person with type O blood

c. Rolling a die and getting an odd number
Rolling a die and getting a number less than 3

d. Randomly selecting a person who is under 21 years of age
Randomly selecting a person who is over 30 years of age

Example 13: Mutually exclusive events

SOLUTION

a. These events are not mutually exclusive since a student can be both female and a junior.

b. These events are mutually exclusive since a person cannot have type A blood and type O blood at the same time.

c. These events are not mutually exclusive since the number 1 is both an odd number and a number less than 3.

d. These events are mutually exclusive since a person cannot be both under 21 and over 30 years of age at the same time.

Example 14: Mutually exclusive events

Determine which events are mutually exclusive and which events are not when a single card is drawn from a deck.

a. Getting a king; getting a diamond

b. Getting a 4; getting a king

c. Getting a face card; getting a club

d. Getting a face card; getting a 10

SOLUTION

a. These events are not mutually exclusive since the king of diamonds represents both events.

b. These events are mutually exclusive since you cannot draw one card that is both a 4 and a king.

c. These events are not mutually exclusive since a jack, queen, or king can be clubs.

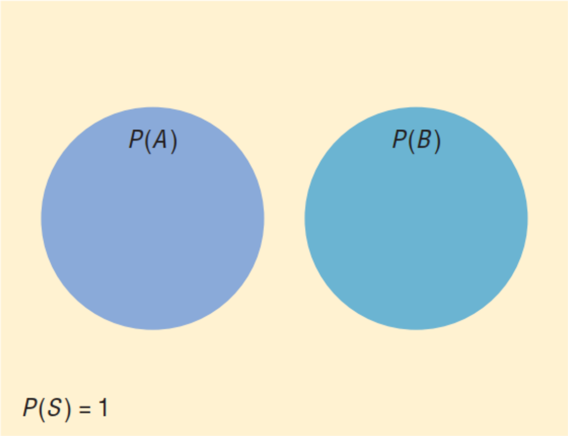
d. These events are mutually exclusive since a 10 and a face card cannot be drawn at the same time when one card is drawn.

Addition rules

Addition Rule 1

When two events A and B are mutually exclusive, the probability that A or B will occur is

$$P(A \text{ or } B) = P(A) + P(B)$$



Mutually exclusive events
$P(A \text{ or } B) = P(A) + P(B)$

The probability rules can be extended to three or more events. For three mutually exclusive events A, B, and C,

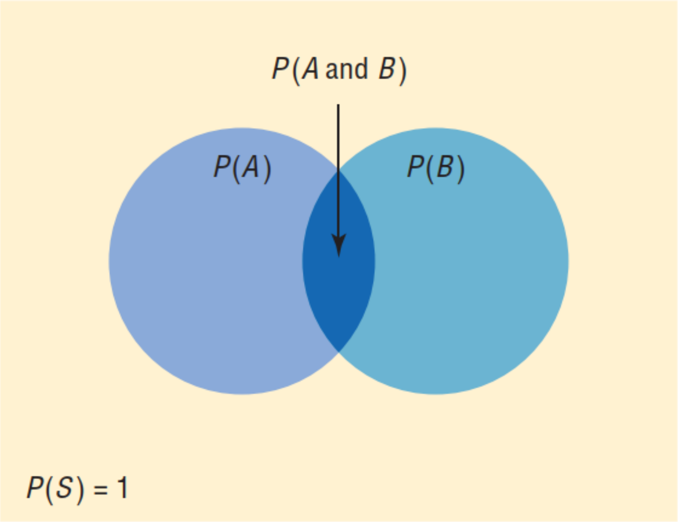
$$P(A \text{ or } B \text{ or } C) = P(A) + P(B) + P(C)$$

Addition rules

Addition Rule 2

If A and B are *not* mutually exclusive, then

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

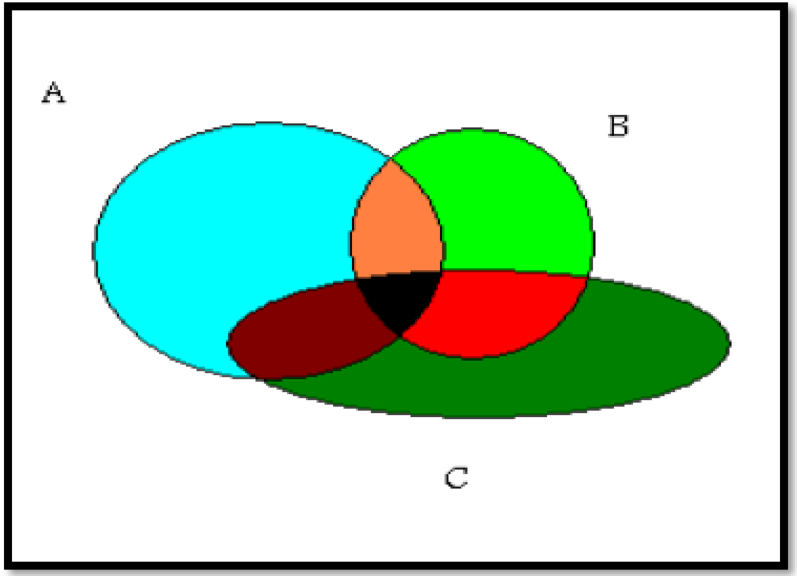


Nonmutually exclusive events
$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

Addition rules

For three events that are *not* mutually exclusive,

$$P(A \text{ or } B \text{ or } C) = P(A) + P(B) + P(C) - P(A \text{ and } B) - P(A \text{ and } C) - P(B \text{ and } C) + P(A \text{ and } B \text{ and } C)$$



Examples 15, 16: Addition rules

A city has 9 coffee shops: 3 Starbucks, 2 Caribou Coffees, and 4 Crazy Mocho Coffees. If a person selects one shop at random to buy a cup of coffee, find the probability that it is either a Starbucks or Crazy Mocho Coffees.

SOLUTION

Since there are 3 Starbucks and 4 Crazy Mochos, and a total of 9 coffee shops, $P(\text{Starbucks or Crazy Mocho}) = P(\text{Starbucks}) + P(\text{Crazy Mocho}) = \frac{3}{9} + \frac{4}{9} = \frac{7}{9} \approx 0.778$. The events are mutually exclusive.

The corporate research and development centers for three local companies have the following numbers of employees:

U.S. Steel	110
Alcoa	750
Bayer Material Science	250

If a research employee is selected at random, find the probability that the employee is employed by U.S. Steel or Alcoa.

SOLUTION

$$P(\text{U.S. Steel or Alcoa}) = P(\text{U.S. Steel}) + P(\text{Alcoa})$$

$$= \frac{110}{1110} + \frac{750}{1110} = \frac{860}{1110} = \frac{86}{111}$$

Example 17: Addition rules

In a survey, 8% of the respondents said that their favorite ice cream flavor is cookies and cream, and 6% like mint chocolate chip. If a person is selected at random, find the probability that her or his favorite ice cream flavor is either cookies and cream or mint chocolate chip.

SOLUTION

$$\begin{aligned} &P(\text{cookies and cream or mint chocolate chip}) \\ &\quad = P(\text{cookies and cream}) + P(\text{mint chocolate chip}) \\ &\quad = 0.08 + 0.06 = 0.14 = 14\% \end{aligned}$$

These events are mutually exclusive.

Example 18: Addition rules

A single card is drawn at random from an ordinary deck of cards. Find the probability that it is either an ace or a black card.

SOLUTION

Since there are 4 aces and 26 black cards (13 spades and 13 clubs), 2 of the aces are black cards, namely, the ace of spades and the ace of clubs. Hence, the probabilities of the two outcomes must be subtracted since they have been counted twice.

$$P(\text{ace or black card}) = P(\text{ace}) + P(\text{black card}) - P(\text{black aces})$$
$$= \frac{4}{52} + \frac{26}{52} - \frac{2}{52} = \frac{28}{52} = \frac{7}{13} \approx 0.538$$

Example 19: Addition rules

In a hospital unit there are 8 nurses and 5 physicians; 7 nurses and 3 physicians are females. If a staff person is selected, find the probability that the subject is a nurse or a male.

SOLUTION

The sample space is shown here.

Staff	Females	Males	Total
Nurses	7	1	8
Physicians	3	2	5
Total	10	3	13

The probability is

$$P(\text{nurse or male}) = P(\text{nurse}) + P(\text{male}) - P(\text{male nurse})$$

$$= \frac{8}{13} + \frac{3}{13} - \frac{1}{13} = \frac{10}{13} \approx 0.769$$

Multiplication rules

Two events *A* and *B* are **independent events** if the fact that *A* occurs does not affect the probability of *B* occurring.

Multiplication Rule 1

When two events are independent, the probability of both occurring is

$$P(A \text{ and } B) = P(A) \cdot P(B)$$

EXAMPLE 20

A coin is flipped and a die is rolled. Find the probability of getting a head on the coin and a 4 on the die.

SOLUTION

The sample space for the coin is H, T; and for the die it is 1, 2, 3, 4, 5, 6.

$$P(\text{head and } 4) = P(\text{head}) \cdot P(4) = \frac{1}{2} \cdot \frac{1}{6} = \frac{1}{12} \approx 0.083$$

Example 21: Multiplication rules

A card is drawn from a deck and replaced; then a second card is drawn. Find the probability of getting a queen and then an ace.

SOLUTION

The probability of getting a queen is $\frac{4}{52}$, and since the card is replaced, the probability of getting an ace is $\frac{4}{52}$. Hence, the probability of getting a queen and an ace is

$$P(\text{queen and ace}) = P(\text{queen}) \cdot P(\text{ace}) = \frac{4}{52} \cdot \frac{4}{52} = \frac{16}{2704} = \frac{1}{169} \approx 0.006$$

Example 22: Multiplication rules

An urn contains 3 red balls, 2 blue balls, and 5 white balls. A ball is selected and its color noted. Then it is replaced. A second ball is selected and its color noted. Find the probability of each of these.

a. Selecting 2 blue balls

b. Selecting 1 blue ball and then 1 white ball

c. Selecting 1 red ball and then 1 blue ball

SOLUTION

a. $P(\text{blue and blue}) = P(\text{blue}) \cdot P(\text{blue}) = \frac{2}{10} \cdot \frac{2}{10} = \frac{4}{100} = \frac{1}{25} = 0.04$

b. $P(\text{blue and white}) = P(\text{blue}) \cdot P(\text{white}) = \frac{2}{10} \cdot \frac{5}{10} = \frac{10}{100} = \frac{1}{10} = 0.1$

c. $P(\text{red and blue}) = P(\text{red}) \cdot P(\text{blue}) = \frac{3}{10} \cdot \frac{2}{10} = \frac{6}{100} = \frac{3}{50} = 0.06$

Example 23: Multiplication rules

Multiplication rule 1 can be extended to three or more independent events by using the formula

$$P(A \text{ and } B \text{ and } C \text{ and } \ldots \text{ and } K) = P(A) \cdot P(B) \cdot P(C) \cdots P(K)$$

A Harris poll found that 46% of Americans say they suffer great stress at least once a week. If three people are selected at random, find the probability that all three will say that they suffer great stress at least once a week.

SOLUTION

Let S denote stress. Then

$$\begin{aligned} P(S \text{ and } S \text{ and } S) &= P(S) \cdot P(S) \cdot P(S) \\ &= (0.46)(0.46)(0.46) \approx 0.097 \end{aligned}$$

There is a 9.7% chance that all three people will say they suffer great stress at least once a week.

Example 24: Multiplication rules

Approximately 9% of men have a type of color blindness that prevents them from distinguishing between red and green. If 3 men are selected at random, find the probability that all of them will have this type of red-green color blindness.

SOLUTION

Let C denote red-green color blindness. Then

$$\begin{aligned} P(C \text{ and } C \text{ and } C) &= P(C) \cdot P(C) \cdot P(C) \\ &= (0.09)(0.09)(0.09) \\ &= 0.000729 \end{aligned}$$

Hence, the rounded probability is 0.0007.

There is a 0.07% chance that all three men selected will have this type of red-green color blindness.

Dependent events

When the outcome or occurrence of the first event affects the outcome or occurrence of the second event in such a way that the probability is changed, the events are said to be **dependent events**.

Here are some examples of dependent events:

Drawing a card from a deck, not replacing it, and then drawing a second card

Selecting a ball from an urn, not replacing it, and then selecting a second ball

Being a lifeguard and getting a suntan

Having high grades and getting a scholarship

Parking in a no-parking zone and getting a parking ticket

Multiplication rules: Conditional probability

The **conditional probability** of an event B in relationship to an event A is the probability that event B occurs after event A has already occurred. The notation for conditional probability is $P(B|A)$. This notation does not mean that B is divided by A; rather, it means the probability that event B occurs given that event A has already occurred.

Multiplication Rule 2

When two events are dependent, the probability of both occurring is

$$P(A \text{ and } B) = P(A) \cdot P(B|A)$$

Formula for Conditional Probability

The probability that the second event B occurs given that the first event A has occurred can be found by dividing the probability that both events occurred by the probability that the first event has occurred. The formula is

$$P(B|A) = \frac{P(A \text{ and } B)}{P(A)}$$

Examples 25, 26: Multiplication rules and conditional probability

In a recent survey, 33% of the respondents said that they feel that they are overqualified (O) for their present job. Of these, 24% said that they were looking for a new job (J). If a person is selected at random, find the probability that the person feels that he or she is overqualified and is also looking for a new job.

SOLUTION

$$P(\text{O and J}) = P(\text{O}) \cdot P(\text{J}|\text{O}) = (0.33)(0.24) \approx 0.079$$

There is about a 7.9% probability that the person feels that he or she is overqualified and is also looking for a new job.

World Wide Insurance Company found that 53% of the residents of a city had homeowner's insurance (H) with the company. Of these clients, 27% also had automobile insurance (A) with the company. If a resident is selected at random, find the probability that the resident has both homeowner's and automobile insurance with World Wide Insurance Company.

SOLUTION

$$P(\text{H and A}) = P(\text{H}) \cdot P(\text{A}|\text{H}) = (0.53)(0.27) = 0.1431 \approx 0.143$$

There is about a 14.3% probability that a resident has both homeowner's and automobile insurance with World Wide Insurance Company.

Example 27: Multiplication rules and conditional probability

Three cards are drawn from an ordinary deck and not replaced. Find the probability of these events.

a. Getting 3 jacks

b. Getting an ace, a king, and a queen in order

c. Getting a club, a spade, and a heart in order

d. Getting 3 clubs

SOLUTION

a. $P(\text{3 jacks}) = \frac{4}{52} \cdot \frac{3}{51} \cdot \frac{2}{50} = \frac{24}{132{,}600} = \frac{1}{5525} \approx 0.0002$

b. $P(\text{ace and king and queen}) = \frac{4}{52} \cdot \frac{4}{51} \cdot \frac{4}{50} = \frac{64}{132{,}600} = \frac{8}{16{,}575} \approx 0.0005$

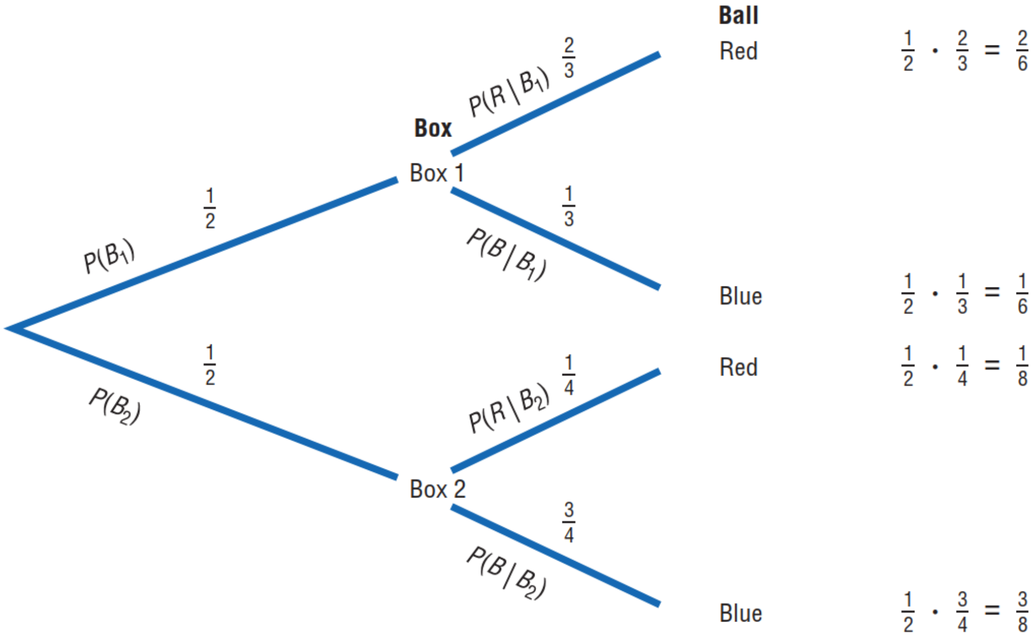
c. $P(\text{club and spade and heart}) = \frac{13}{52} \cdot \frac{13}{51} \cdot \frac{13}{50} = \frac{2197}{132{,}600} = \frac{169}{10{,}200} \approx 0.017$

d. $P(\text{3 clubs}) = \frac{13}{52} \cdot \frac{12}{51} \cdot \frac{11}{50} = \frac{1716}{132{,}600} = \frac{11}{850} \approx 0.013$

Example 28: Multiplication rules and conditional probability

Box 1 contains 2 red balls and 1 blue ball. Box 2 contains 3 blue balls and 1 red ball. A coin is tossed. If it falls heads up, box 1 is selected and a ball is drawn. If it falls tails up, box 2 is selected and a ball is drawn. Find the probability of selecting a red ball.

SOLUTION



Example 28: Multiplication rules and conditional probability

SOLUTION

multiply the probability for each outcome, using the rule $P(A \text{ and } B) = P(A) \cdot P(B|A)$. For example, the probability of selecting box 1 and selecting a red ball is $\frac{1}{2} \cdot \frac{2}{3} = \frac{2}{6}$. The probability of selecting box 1 and a blue ball is $\frac{1}{2} \cdot \frac{1}{3} = \frac{1}{6}$. The probability of selecting box 2 and selecting a red ball is $\frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$. The probability of selecting box 2 and a blue ball is $\frac{1}{2} \cdot \frac{3}{4} = \frac{3}{8}$. (Note that the sum of these probabilities is 1.)

a red ball can be selected from either box 1 or box 2 so $P(\text{red}) = \frac{2}{6} + \frac{1}{8} = \frac{8}{24} + \frac{3}{24} = \frac{11}{24}$.

Example 29: Multiplication rules and conditional probability

A box contains black chips and white chips. A person selects two chips without replacement. If the probability of selecting a black chip *and* a white chip is $\frac{15}{56}$ and the probability of selecting a black chip on the first draw is $\frac{3}{8}$, find the probability of selecting the white chip on the second draw, *given* that the first chip selected was a black chip.

SOLUTION

Let

$$B = \text{selecting a black chip} \qquad W = \text{selecting a white chip}$$

Then

$$P(W|B) = \frac{P(B \text{ and } W)}{P(B)} = \frac{15/56}{3/8}$$

$$= \frac{15}{56} \div \frac{3}{8} = \frac{15}{56} \cdot \frac{8}{3} = \frac{\overset{5}{\cancel{15}}}{\underset{7}{\cancel{56}}} \cdot \frac{\overset{1}{\cancel{8}}}{\underset{1}{\cancel{3}}} = \frac{5}{7} \approx 0.714$$

Hence, the probability of selecting a white chip on the second draw given that the first chip selected was black is $\frac{5}{7} \approx 0.714$.

Example 30: Multiplication rules and conditional probability

The probability that Sam parks in a no-parking zone *and* gets a parking ticket is 0.06, and the probability that Sam cannot find a legal parking space and has to park in the no-parking zone is 0.20. On Tuesday, Sam arrives at school and has to park in a no-parking zone. Find the probability that he will get a parking ticket.

SOLUTION

Let

$$N = \text{parking in a no-parking zone} \qquad T = \text{getting a ticket}$$

Then

$$P(T|N) = \frac{P(N \text{ and } T)}{P(N)} = \frac{0.06}{0.20} = 0.30$$

Hence, Sam has a 0.30 probability or 30% chance of getting a parking ticket, given that he parked in a no-parking zone.

Example 31: Multiplication rules and conditional probability

A person selects 3 cards from an ordinary deck and replaces each card after it is drawn. Find the probability that the person will get at least one heart.

SOLUTION

Let

$$E = \text{at least 1 heart is drawn} \qquad \text{and} \qquad \overline{E} = \text{no hearts are drawn}$$

$$P(\overline{E}) = \frac{39}{52} \cdot \frac{39}{52} \cdot \frac{39}{52} = \frac{3}{4} \cdot \frac{3}{4} \cdot \frac{3}{4} = \frac{27}{64}$$

$$\begin{aligned} P(E) &= 1 - P(\overline{E}) \\ &= 1 - \frac{27}{64} = \frac{37}{64} \approx 0.578 = 57.8\% \end{aligned}$$

Hence, a person will select at least one heart about 57.8% of the time.

Example 32: Multiplication rules and conditional probability

A coin is tossed 5 times. Find the probability of getting at least 1 tail.

SOLUTION

It is easier to find the probability of the complement of the event, which is "all heads," and then subtract the probability from 1 to get the probability of at least 1 tail.

$$P(E) = 1 - P(\bar{E})$$

$$P(\text{at least 1 tail}) = 1 - P(\text{all heads})$$

$$P(\text{all heads}) = \left(\frac{1}{2}\right)^5 = \frac{1}{32}$$

Hence,

$$P(\text{at least 1 tail}) = 1 - \frac{1}{32} = \frac{31}{32} \approx 0.969$$

There is a 96.9% chance of getting at least one tail.

Example 33: Multiplication rules and conditional probability

The Neckware Association of America reported that 3% of ties sold in the United States are bow ties. If 4 customers who purchased a tie are randomly selected, find the probability that at least 1 purchased a bow tie.

SOLUTION

Let E = the tie purchased is a bow tie and $\overline{E}$ = the tie purchased is not a bow tie. Then

$$P(E) = 0.03 \qquad \text{and} \qquad P(\overline{E}) = 1 - 0.03 = 0.97$$

$$P(\text{no bow ties are purchased}) = (0.97)(0.97)(0.97)(0.97) \approx 0.885; \text{ hence,}$$
$$P(\text{at least one bow tie is purchased}) = 1 - 0.885 = 0.115.$$

There is an 11.5% chance of a person purchasing at least one bow tie.